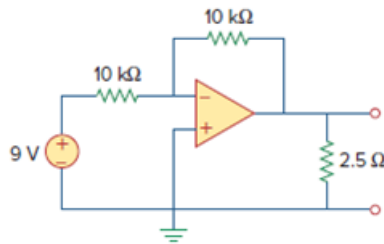


Problem

For the circuit shown in Figure 5.57, solve for the Thevenin equivalent circuit looking into terminals A and B.

Figure 5.57:



Step-by-step solution

Step 1 of 6

Refer to Figure 5.57 in the textbook.

Thevenin's theorem:

The linear two-terminal circuit is replaced by a single voltage source in series with the resistor. Here, the single voltage source is called the Thevenin's voltage and series resistance is called the Thevenin's resistance.

The modified circuit is shown in Figure 1 to calculate the Thevenin's voltage.

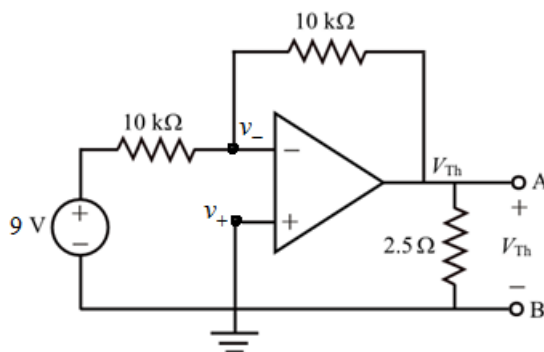


Figure 1

Step 2 of 6

From Figure 1, observe that the circuit is an inverting amplifier; since the input voltage is applied to the inverting terminal. The non-inverting terminal is connected to the ground. According to the op-amp virtual ground concept, the voltage at inverting and non-inverting terminal is same.

Apply the Kirchhoff's current law at inverting terminal of the op-amp.

$$\begin{aligned}\frac{v_- - 9}{10 \text{ k}} + \frac{v_- - V_{Th}}{10 \text{ k}} &= 0 \\ \frac{9}{10 \text{ k}} + \frac{V_{Th}}{10 \text{ k}} &= 0 \quad (\text{since } v_- = v_+ = 0) \\ 9 + V_{Th} &= 0 \\ V_{Th} &= -9 \text{ V}\end{aligned}$$

Therefore, the Thevenin's voltage is $\boxed{-9 \text{ V}}$.

Step 3 of 6

The Thevenin's voltage is an open circuit voltage (Note: The op-amp keeps the output voltage at -9 volts in spite of any connection between A and B). This means that even a short from A to B would theoretically produce an infinite current.

Refer to Figure 5.4 and 5.11 in the textbook. For an ideal op-amp input resistance is infinite and the output resistance is zero.

Re-draw the circuit diagram considering the equivalent circuit of op-amp.

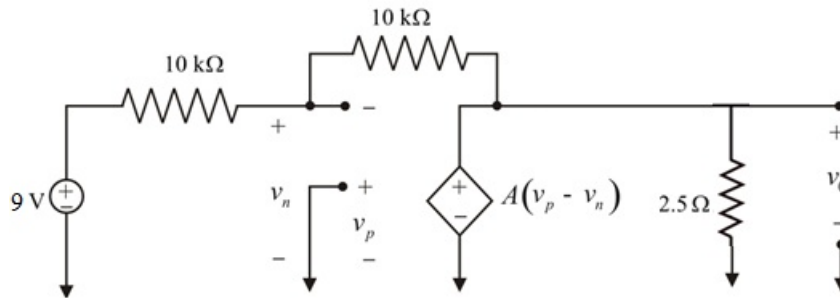


Figure 2

Step 4 of 6

Determine the Thevenin's resistance.

Replace all the voltage source with their internal resistances.

Re-draw the circuit diagram.

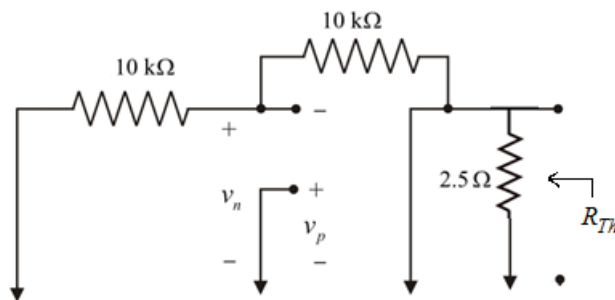


Figure 3

Step 5 of 6

From Figure 3, determine the Thevenin's resistance.

$$R_{Th} = \frac{(0)(2.5)}{0 + 2.5}$$

$$= 0 \Omega$$

Therefore, the Thevenin's resistance is 0Ω .

Step 6 of 6

Draw the Thevenin's equivalent circuit.

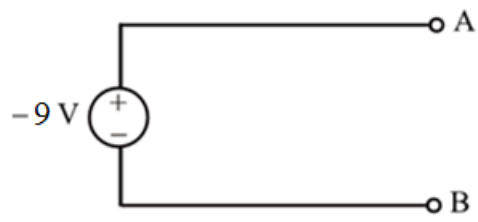


Figure 4

Therefore, the Thevenin's equivalent circuit is shown in Figure 4.